



CANDIDATE
NAME

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CENTRE
NUMBER

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CANDIDATE
NUMBER

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0625/42

Practice Paper

1 hour 15 minutes

No additional materials are needed.

- Answer **all** questions.
- Use a black or dark blue pen. You may use an HB pencil for any diagrams or graphs.
- Write your name, centre number and candidate number in the boxes at the top of the page.
- Write your answer to each question in the space provided.
- Do **not** use an erasable pen or correction fluid.
- Do **not** write on any bar codes.
- You may use a calculator.
- You should show all your working and use appropriate units.
- Take the weight of 1.0 kg to be 10 N (acceleration of free fall = 10 m/s^2).

- The total mark for this paper is 80.
- The number of marks for each question or part question is shown in brackets [].

DC (PRACTICE) 000000/1

[Turn over

- 1 Fig. 1.1 shows an experimental arrangement used to investigate energy transfer.

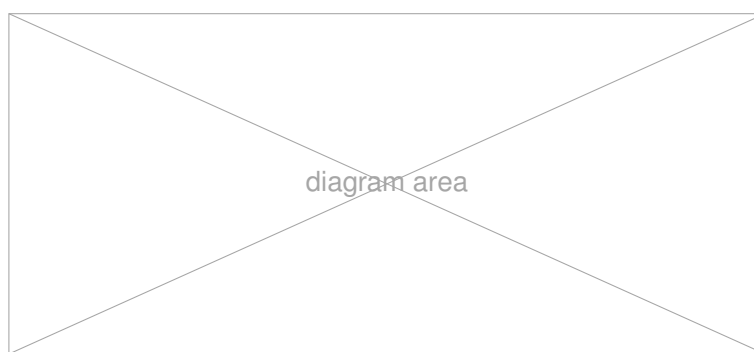


Fig. 1.1

The apparatus transfers energy to a small load.

- (a) (i) State the form of useful energy gained by the load.

..... [1]

- (ii) The load has a mass of 0.45 kg and is raised through a vertical height of 1.8 m.

Calculate the useful energy transferred to the load.

energy = J [3]

- (b) The input energy supplied to the motor is 12 J.

Calculate the efficiency of the motor.

efficiency = % [3]

- (c) State one reason why the efficiency is less than 100%.

..... [1]

[Total: 8]

- 2 A trolley is released from rest and moves down a slope.

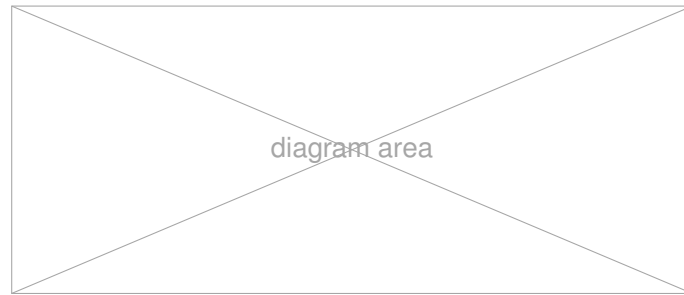


Fig. 2.1

- (a) Define acceleration.

.....

[2]

- (b) (i) The trolley reaches a speed of 3.6 m/s after 1.2 s. Calculate its acceleration.

acceleration = m/s² [3]

- (ii) The mass of the trolley is 1.5 kg. Calculate the resultant force acting on the trolley.

force = N [2]

- (c) Suggest two changes that would reduce the acceleration of the trolley.

1.

2.

[2]

[Total: 9]

- 3 A student investigates the resistance of a wire.

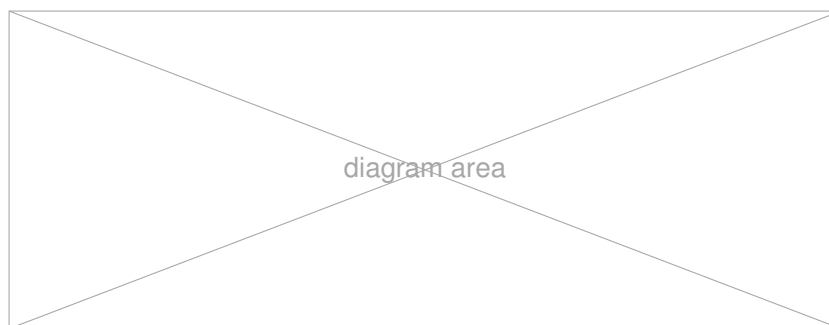


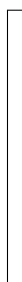
Fig. 3.1

- (a) Complete the table headings for the measurements needed in this experiment.

Length of wire / cm		
20		
40		
60		
80		

[2]

- (b) Describe how the student can use the measurements to determine the resistance of the wire.



[4]

- (c) State two precautions that improve the reliability of the results.

1.
2.

[2]

- (d) On the grid, sketch the expected shape of a graph of resistance against length for a uniform wire.



[2]

[Total: 10]

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- 4 This page shows the structure for adding another long question. Replace this text with your own question stem.

(a) Write the first part of the question here.

answer = unit [2]

(b) Write the second part of the question here.

[4]

[Total: 6]